# An adaptive demodulation for OFDM Signal

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***Abstract*—This paper presents a novel method for OFDM demodulation based on LMS adaptive filter. In this proposed method, data symbol is reconstructed from amplitude of reference signal without using fast Fourier transform and thus this method can be applied for real time system. In addition, the convergence rate of adaptive filter can be promoted by using the proper initial weights. The proposed technique will be applied for BASK-OFDM, BPSK-OFDM and QAM-OFDM demodulation. The obtained simulation results of the proposed method compared to the conventional scheme show that the proposed technique can be applied for real time processing of OFDM demodulation.**

***Keywords—adaptive filter; LMS algorithm; initial weight; OFDM; demodulation***

## I. INTRODUCTION

Nowadays, the amenities in communication technology are required such as multimedia mobile communication, high speed internet, video conference and public digital broadcasting. These communications on high data rate digital communication system requires wide bandwidth [1]. Then various techniques are developed to achieve high efficiency of bandwidth usage. Orthogonal Frequency Division Multiplex (OFDM) is one technique which can support high data rate communication [2].

OFDM technique provides more efficient bandwidth usage compared with the same number of subcarrier conventional frequency division multiplexing (FDM) [3]. The data transfer by using OFDM is developed by the ideas of parallel data transmission [4]. It is based on dividing the data into subgroup, each subgroup is modulated with a difference subcarrier which is orthogonal to each other. So, the overlap (aliasing) is passable. Since the original OFDM was presented [5]-[6], many researches have been continuing launched. In addition, after a technique by using inverse discrete Fourier transform (IDFT) for modulation and discrete Fourier transform (DFT) for demodulation OFDM signal was proposed [7]-[8], OFDM has become a standard of transmission in many communication systems ever since. Later, IDFT and DFT can be reduced their mathematical operation by using IFFT and FFT, respectively [9]-[10]. Among these developing OFDM researches, most researchers focus on improving the performance of OFDM system. For example, the using of wavelet coefficients instead of Fourier coefficients was introduced by [11], while a dividing block data to a small subblock for improving data rate of transmission and reducing mathematic operation was presented by [12].

In present, the adaptive filter is employed for demodulation in many researches such as an adaptive synchronous amplitude demodulation technique proposed by Fathy F. Yassa *et al.* [13]. Later, Michael J. Werter presented FSK demodulation using an adaptive recursive digital filter [14] and L. Yanxin and H. Aiqun proposed an adaptive demodulation method for QAM signal [15].

In this paper, a new technique for demodulation OFDM signal is proposed. It is based on using LMS adaptive filter whose initial condition of weight is defined according to the technique proposed in [16]. By using the proposed method, the FFT for reconstruction OFDM signal is not required. The OFDM demodulation using the proposed technique can rapidly reconstructing data symbol and working in real-time system.

The organization of this paper is as follows. Section 2 mentions the principle of OFDM and the OFDM demodulation using adaptive filter. The computer simulation results are shown in section 3. Finally, section 4 discusses the conclusion.

## II. PRINCIPLE AND METHOD

### A. Orthogonal property of OFDM system

Generally, the subcarrier of OFDM system is orthogonal to each other. This technique, a signal in each channel is generated without interference. By using a Fourier transform, which is

$$F(\omega)=\int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} x(t)e^{-j\omega t}dt \qquad (1)$$

where $x(t)$ is the analogue signal whose amplitude is for example, A volts for symbol "1" or –A volts for symbol "0", during the symbol period $T_s$ second. The result of integrating (1) is given by

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$$F(\omega) = AT_s \left( \frac{\sin\left(\frac{T_s\omega}{2}\right)}{\frac{T_s\omega}{2}} \right) = AT_s \,\mathrm{sinc}\left(\frac{T_s\omega}{2}\right) \qquad (2)$$

To obtain the orthogonal property of the sinc function $F(\omega)$, it is defined that $\frac{T_s\omega}{2} = \pi n$ where $n = 1, 2, 3, \ldots$ which results in $F(\omega)$ will be zero when

$$\omega = \frac{2\pi n}{T_s} \qquad (3)$$

The selection of $\omega$ depends on symbol period or data rate. Thus, carrier frequency arrangement can be provided by

$$f_{cn} = f_0 + \frac{n}{T_s} \qquad (4)$$

where $f_0$ is the minimum subcarrier frequency using in the OFDM system.

Fig. 1 illustrates the frequency spectrum of orthogonal subcarriers using (4). From Fig. 1, the peak of each subcarrier is not interfered to each other. The conventional block diagram of OFDM is illustrated in Fig. 2.

From the block diagram shown in Fig. 2(a), the input digital data is divided to $N$ subgroups and demodulated with $N$ subcarriers. The digital data rate is $m$ bit/second, therefore, the data rate of each channel is $\frac{m}{N}$ bit/second. The digital data is transformed to the data symbol, and then the real part of OFDM signal ( $s(n)$ ) by using IFFT will be transmitted. According to Fig. 2(b), $s(n)$ signal is transformed to $d_k$ by using FFT, then the digital data can be reconstructed from $d_k$ .

The OFDM signal in the form of continuous time is expressed in the following equation:

$$s(t) = \mathrm{Re}\left[ \sum_{k=0}^{N-1} d_k e^{j2\pi \frac{k}{T_s} t} \right] \qquad (5)$$

From (5), if the OFDM signal is passed through the sampling process whose sampling time $\frac{T_s}{N}$, it means that there are $N$ samples in $T_s$ seconds. Hence, the OFDM signal in the form of discrete time can be written as

$$s(n) = \mathrm{Re}\left[ \sum_{k=0}^{N-1} d_k e^{j2\pi \frac{k}{N} n} \right]. \qquad (6)$$

And (6) is normalized with $\frac{1}{N}$, it yields

$$s(n) = \frac{1}{N} \mathrm{Re}\left[ \sum_{k=0}^{N-1} d_k e^{j2\pi \frac{k}{N} n} \right] \qquad (7)$$

This equation is the IDFT equation. From (7), $d_k$ is coefficient of the exponential function. Therefore, the data symbol can be reconstructed by using DFT which is

$$d_k = \sum_{n=0}^{N-1} s(n) e^{-j2\pi \frac{k}{N} n} \qquad (8)$$

Fourier transform is the process of transformation the signal in time domain into frequency domain. It is related to the complex function which is

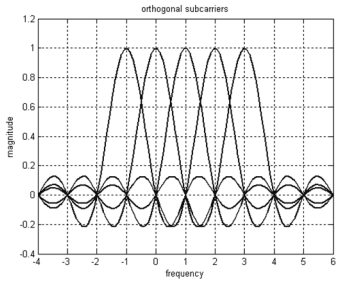
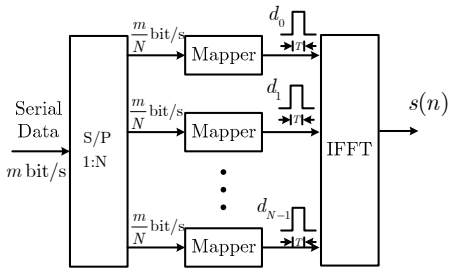
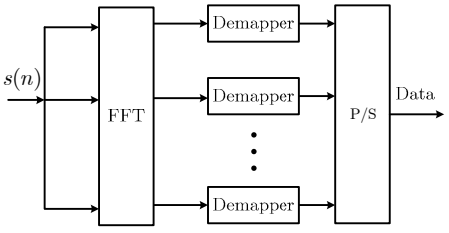



Fig. 1. Spectrum of orthogonal subcarrier in OFDM system




a) OFDM transmission using IFFT




b) OFDM received using FFT

Fig. 2. The conventional block diagram of OFDM system

$$e^{\pm j\omega n} = \cos(\omega n) \pm j\sin(\omega n) \quad (9)$$

where in term of $\cos(\omega n)$ is real part and $\sin(\omega n)$ is imaginary part. In present, OFDM system employs inverse fast Fourier transform (IFFT) and fast Fourier transform (FFT) for modulation and demodulation, respectively.

*B. OFDM demodulation by using adaptive filter*

In this subsection, a new method for OFDM demodulation by using adaptive filter is explained. The block diagram of the proposed adaptive filter for OFDM demodulation is shown in Fig. 3. The output of the system $e(n)$ is

$$e(n) = s(n) - \hat{x}(n) \quad (10)$$

$$\hat{x}(n) = \sum_{k=0}^{N-1} a_k \cos(2\pi \frac{k}{N} n) + \sum_{k=0}^{N-1} b_k \sin(2\pi \frac{k}{N} n) \quad (11)$$

where $s(n)$ is OFDM signal, $\hat{x}(n)$ is reference carrier signal of the system, $a_k, b_k$ are adapted weight that is amplitude of cosine and sine signal, respectively. In (6), the data symbol $d_k$ is the complex number that is

$$d_k = A_k e^{j\phi_k} = I_k + jQ_k \quad , \quad k = 0,1,...,N-1 \quad (12)$$

By using (9), the variables of (12) are given by

$$I_k = A_k \cos\phi_k, Q_k = A_k \sin\phi_k \quad (13)$$

where $I_k$ is inphase component of data symbol, $Q_k$ is quadrature component of data symbol and $\phi_k$ is angle of data symbol.

From (12) and (13), (6) can be rewritten as

$$\begin{aligned} s(n) &= \mathrm{Re}\left[\sum_{k=0}^{N-1} d_k e^{j2\pi\frac{k}{N}n}\right] \\ &= \mathrm{Re}\left[\sum_{k=0}^{N-1} \left(A_k \cos\phi_k + jA_k \sin\phi_k\right) \right. \\ &\qquad \left. \left(\cos\left(2\pi\frac{k}{N}n\right) + j\sin\left(2\pi\frac{k}{N}n\right)\right)\right] \\ &= \sum_{k=0}^{N-1}\left(A_k \cos\phi_k \cos\left(2\pi\frac{k}{N}n\right)\right. \\ &\qquad \left. + A_k \sin\phi_k \sin\left(2\pi\frac{k}{N}n\right)\right) \\ s(n) &= \sum_{k=0}^{N-1} I_k \cos\left(2\pi\frac{k}{N}n\right) + \sum_{k=0}^{N-1}(-Q_k)\sin\left(2\pi\frac{k}{N}n\right). \end{aligned} \quad (14)$$

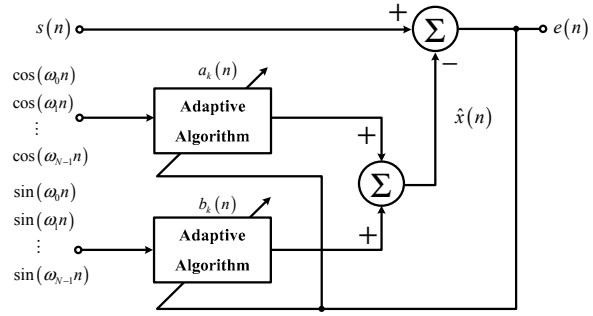



Fig. 3. Block diagram of the proposed OFDM demodulation

By replacing (11) and (14) into (10), it yields

$$\begin{aligned} e(n) &= \sum_{k=0}^{N-1} I_k \cos\left(2\pi\frac{k}{N}n\right) + \sum_{k=0}^{N-1}(-Q_k)\sin\left(2\pi\frac{k}{N}n\right) \\ &\quad - \sum_{k=0}^{N-1} a_k \cos(2\pi\frac{k}{N}n) - \sum_{k=0}^{N-1} b_k \sin(2\pi\frac{k}{N}n) \end{aligned} \quad (15)$$

By considering (15) at the convergent state where $e(n) = 0$

$$\begin{aligned} &\sum_{k=0}^{N-1} I_k \cos\left(2\pi\frac{k}{N}n\right) + \sum_{k=0}^{N-1}(-Q_k)\sin\left(2\pi\frac{k}{N}n\right) \\ &= \sum_{k=0}^{N-1} a_k \cos\left(2\pi\frac{k}{N}n\right) + \sum_{k=0}^{N-1} b_k \sin\left(2\pi\frac{k}{N}n\right) \end{aligned} \quad (16)$$

From (16), it is seen that $a_k = I_k$ and $b_k = -Q_k$, it thus implies that the data symbol can be demodulated by adaptive filter in the form of adapted weight $a_k, b_k$. The parameters $a_k$ and $b_k$ are adapted according to the following equations:

$$\begin{aligned} a_k(n+1) &= a_k(n) + \mu_a \frac{\partial e^2(n)}{\partial a_k(n)} \\ &= a_k(n) + 2\mu_a e(n)\cos\left(2\pi\frac{k}{N}n\right) \end{aligned} \quad (17)$$

$$\begin{aligned} b_k(n+1) &= b_k(n) - \mu_b \frac{\partial e^2(n)}{\partial b_k(n)} \\ &= b_k(n) + 2\mu_b e(n)\sin\left(2\pi\frac{k}{N}n\right) \end{aligned} \quad (18)$$

where $\mu_a$ , $\mu_b$ are the step size values of the adaptive algorithm, which is $0 < \mu_a, \mu_b < 1$.

## III. The Simulation Results

The simulation results for OFDM demodulation by using adaptive filter with the optimal initial weight are shown in this section. The simulation is obtained by using MATLAB and the

algorithm of adaptive filter is based on least mean square (LMS). The step size parameters are $\mu_a, \mu_b = 0.032$ defined from the intersection point between the graph of misadjustment $(M)$ and the iteration number of weight adjustment process $(\tau)$ [16]. The proposed technique and the conventional technique are used to compare the reconstructed data symbol with noise at SNR = 5 dB. The sample number of 43, 38 and 84 samples are employed to finding the initial weight according to the technique given in [15] for BASK-OFDM, BPSK-OFDM and QAM-OFDM, respectively. For the simulation, it is assumed that 1 symbol has 10 bits so there are 10 subcarriers where the fundamental frequency is $0.01\pi$ radians. The data symbol employed in BASK-OFDM and BPSK-OFDM are shown in Fig. 4 and 8, respectively. For Fig. 12 and 13, the inphase components and quadrature components of QAM-OFDM are depicted.

For demodulating BASK-OFDM signal, Fig. 5 depicts the BASK-OFDM signal with noise at SNR = 5 dB where the reconstructed data symbol is shown in Fig. 6. From the result, the proposed method can reconstruct data symbol more accurate than the conventional method. The behavior of adapted weights of the adaptive filter is depicted in Fig. 7. From Fig. 7, the number of samples, which is used to reach the convergence state after employ the proper the initial weight, is 53 samples. By considering Fig. 7, the proposed method can stop process and reconstruct data symbol after 53 iterations has been processed, whereas using of FFT method must wait for all samples of the symbol in order to start process and also needs all of them to reconstruct data symbol. It implies that the proposed method is in the process to reconstruct the data symbol while the FFT method is still waiting to completely collect all symbols and reconstruction process has not yet been started. Similarly, for demodulating BPSK-OFDM signal and QAM-OFDM signal, the results are demonstrated in Fig. 9-11 and 14-18, sequentially.

## IV. CONCLUSION

This article proposes a new method for OFDM demodulation by using adaptive filter with the proper initial condition. The simulation results show that the proposed technique can reconstruct correct data symbol for BASK-OFDM, BPSK-OFDM and QAM-OFDM compared to the conventional technique for the case with noise at SNR = 5 dB. The adapted weight of adaptive filter shows that the proposed technique can stop adapting process and reconstruct data symbol after the adaptive filter reaches to the convergent state. Contrarily to the conventional technique that all samples of the symbol are required for reconstruction, the proposed technique can reconstruct data symbol after the adaptive filter reaches its convergent state in a concise time. Comparing to the conventional technique, the proposed method can be applied for real time processing due to the properties of adaptive filter. However, it should be noted that in term of mathematical operations, the proposed technique requires more operations than the conventional one.

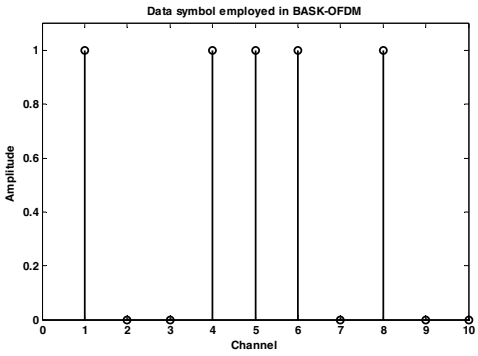



Fig. 4. Data symbol employed in BASK-OFDM

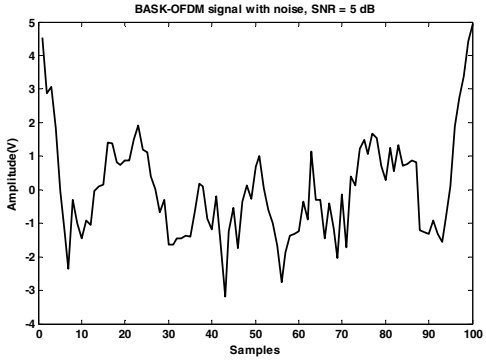



Fig. 5. BASK-OFDM signal with noise at SNR = 5 dB

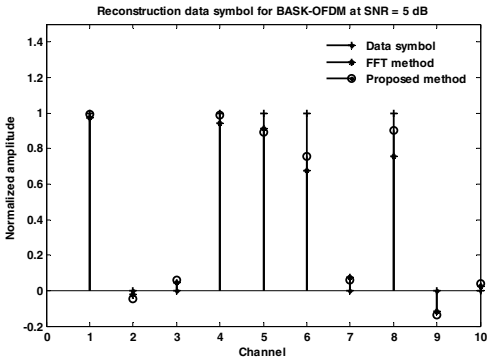



Fig. 6. Reconstructed data symbol
for BASK-OFDM signal with noise at SNR = 5 dB

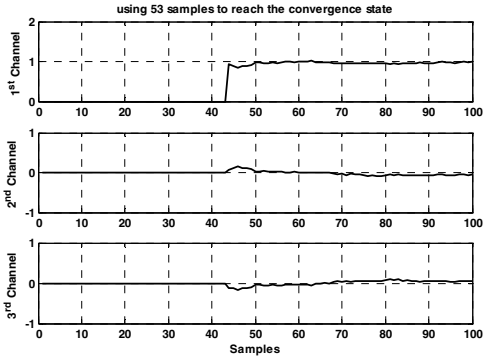



Fig. 7. Adapted weight of adaptive filter for BASK-OFDM

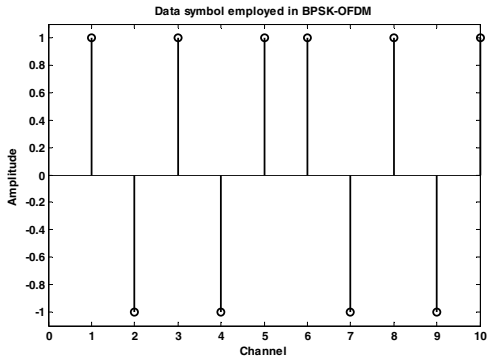



Fig. 8. Data symbol employed in BPSK-OFDM

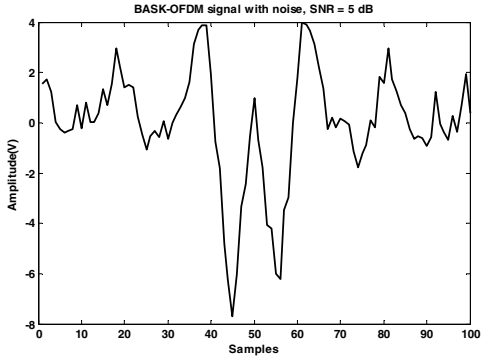



Fig. 9. BPSK-OFDM signal with noise at SNR = 5 dB

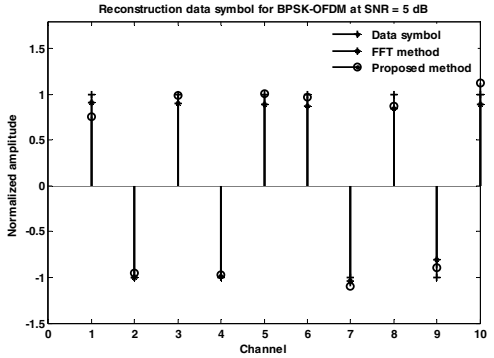



Fig. 10. Reconstructed data symbol
for BPSK-OFDM signal with noise at SNR = 5 dB

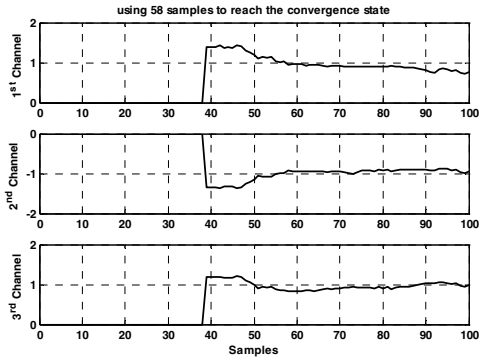



Fig. 11. Adapted weight of adaptive filter for BPSK-OFDM

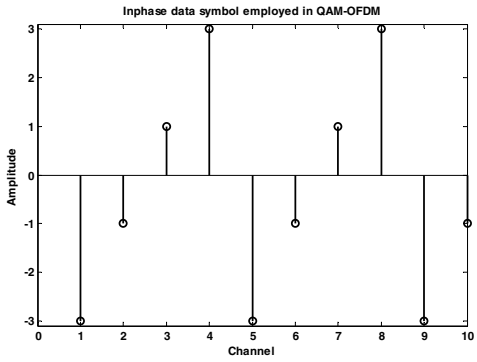



Fig.12. Inphase components in QAM-OFDM

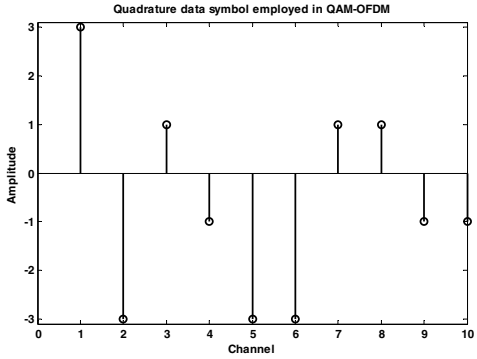



Fig. 13. Quadrature components in QAM-OFDM

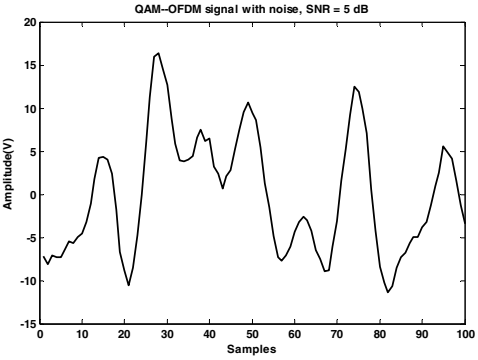



Fig. 14. QAM-OFDM signal with noise at SNR = 5 dB

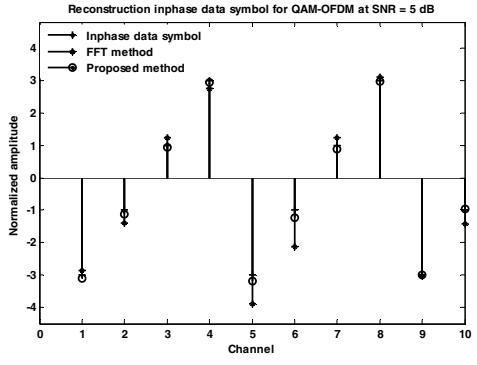



Fig. 15. Reconstructed inphase component
for QAM-OFDM signal with noise at SNR = 5 dB

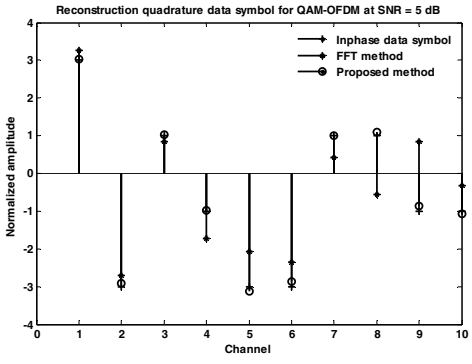



Fig. 16. Reconstructed quadrature component for QAM-OFDM signal with noise at SNR = 5 dB

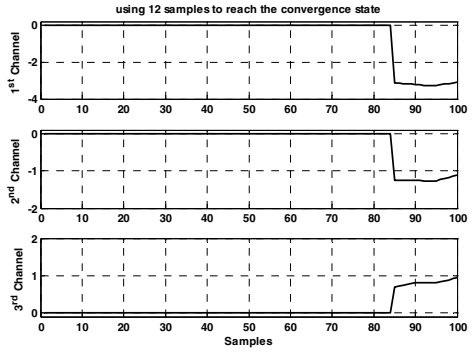



Fig. 17. Adapted inphase component for QAM-OFDM

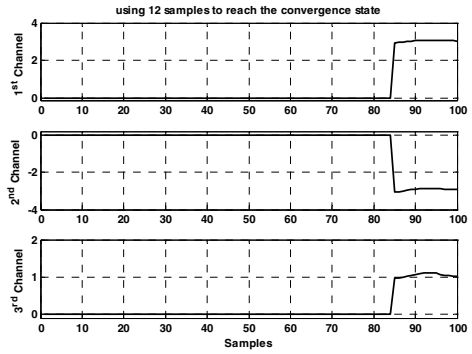



Fig. 18. Adapted quadrature component for QAM-OFDM

## REFERENCES


[1] V.K. Garg, *Wireless Network Evolution 2G to 3G*, NJ: Prentice Hall, 2001.

[2] R.V. Nee and R. Prasad, *OFDM for Wireless Multimedia Communication,* 1st ed., MA: Artech House, 2000.

[3] A.F. Molisch, A. Mamela and D.P. Taylor, *Wideband Wireless digital Communication*, NJ: Prentice Hall, 2000.

[4] R.W. Chang, "Synthesis of Band limited Orthogonal Signals for Multichannel Data Transmission," *Bell System Technical Journal*,vol 45, pp. 1775-1796, 1966.

[5] B.R. Saltzberg, "Performance of an Efficient Parallel Data Transmission System," *IEEE Trans. Commu*., pp. 805-811, 1967.

[6] R.W. Chang and R.A. Gibby, "Theoretical Study of Performance of an Orthogonal Multiplexing Data Transmission Scheme," *IEEE Trans. Commu.*, pp. 529-540, 1968.

[7] S.B. Weinstein and P.M. Ebert, "Data Transmission by Frequency-Division Multiplexing Using the Discrete Fourier Transform," *IEEE Trans. Commu*., pp. 628-634, 1971.

[8] A. Peled, A. Ruiz, "Frequency Domain Data Transmission Using Reduced Computational Complexity Algorithms," *in Proceedings of the IEEE Int. Conf. Acoust., Speech, and Signal Process. (ICASSP'80),* 1980, pp. 964-967.

[9] B. Hirosaki, "An Orthogonally Multiplexed QAM System Using the Discrete Fourier Transform," *IEEE Trans. Commu*., pp. 982-989, 1981

[10] G. Cariolaro, F.C. Vagliani, "An OFDM scheme with a Half Complexity," *IEEE J. Commu.*, pp. 1586 – 1599, 1995.

[11] X. Zhang, G. Bi, "OFDM scheme based on complex orthogonal wavelet packet," *Proc. 12th IEEE Int. Symp. Personal Indoor Mobile Radio Commu.*, pp. E-99 - E-102, 2001.

[12] S.G. Kang, "A novel subblock partition scheme for partial transmit sequence OFDM," *IEEE Trans. Broadcasting*, pp. 333 - 338, 1999.

[13] F. F. Yassa and S. E. Noujaim, "Adaptive Synchronous Amplitude Demodulation", *IEEE Conf. Signals Syst. and Comput.,* vol. 1, pp. 107 - 111, 1988.

[14] M. J. Werter, "FSK demodulation using an adaptive recursive digital filter," *IEEE Trans. Acoust., Speech, and Signal Process.,* vol. 44, no. 7, pp. 1822 - 1825, Jul. 1996.

[15] L. Yanxin and H. Aiqun, "An Adaptive Demodulation Method for QAM Signals", *IEEE Int. Symp. Microwave, Antenna, Propagation and EMC Technologies for Wireless Commu.*, pp. 1064-1067, Aug. 2007.

[16] A. Manosueb, J. Koseeyaporn and P. Wardkein, "PLI cancellation in ECG signal based on adaptive filter by using Wiener-Hopf equation for providing initial condition" *Hindawi J. Computational and Math. Methods in Medicine,* vol. 2014, 2014.